

Multidimensional time-series analysis of ground deformation from multiple InSAR data sets applied to Virunga Volcanic Province

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SUMMARY

A novel, multidimensional small baseline subset (MSBAS) methodology is presented for integration of multiple interferometric synthetic aperture radar (InSAR) data sets for computation of 2- or 3-D time-series of deformation. The proposed approach allows the combination of all possible air-borne and space-borne SAR data acquired with different acquisition parameters, temporal and spatial sampling and resolution, wave-band and polarization. The produced time-series have improved temporal resolution and can be enhanced by applying either regularization or temporal filtering to remove high-frequency noise. We apply this methodology to map 2003–2010 ground deformation of the Virunga Volcanic Province (VVP), North Kivu, Democratic Republic of Congo. The horizontal and vertical time-series of ground displacement clearly identify lava compaction areas, long-term deformation of Mt Nyamuragira and 2004, 2006 and 2010 pre- and co-eruptive deformation. Providing that enough SAR data is available, the method opens new opportunities for detecting ground motion in the VVP and elsewhere.

Key words: Time-series analysis; Numerical solutions; Inverse theory; Satellite geodesy; Radar interferometry; Africa.

1 INTRODUCTION

Many Synthetic Aperture Radar (SAR) images from space-borne SAR sensors, mostly C-band (ERS-1/2, ENVISAT, RADARSAT-1/2) but also X-band (TerraSAR-X, CosmoSkyMed) and L-band (JERS-1, ALOS) have been collected since the launch of the first SAR satellite, ERS-1 in 1992, achieving almost continuous worldwide coverage. Many regions were simultaneously imaged by multiple SAR sensors with varying acquisition parameters (i.e. azimuth and incidence angles, spatial and temporal resolutions, polarization) and various studies utilized InSAR data acquired by different sensors (e.g. Biggs *et al.* 2010; Lu *et al.* 2010; Wei *et al.* 2010; Stramondo *et al.* 2011). Utilization of multiple data sets may be beneficial for cross validation and estimation of the accuracy of InSAR but because each InSAR set is acquired in a particular orbital geometry and temporal sampling, the direct comparison of different sets is complicated if at all possible. For mapping ground deformation over an extended period of time and particularly for multidimensional time-series analysis, it is desirable to utilize all available InSAR data at once, including air-borne and space-borne data from sensors with varying acquisition geometries (e.g. Phase 3 ENVISAT), to compensate for the limitations of a particular data set (e.g. low temporal resolution) and to provide uninterrupted coverage.

The methodology for producing time-series of the line-of-sight deformation from a single set of InSAR observations was proposed by Berardino *et al.* (2002) and Usai (2003) and is presently an established technique known as the small baseline subset (SBAS). Time-series analysis of InSAR data has proven to be useful for mapping various ground deformation, including seismic and volcanic (e.g. Wright *et al.* 2001; Beavan *et al.* 2010) subsidence due to mining and fluid extraction (e.g. Samsonov *et al.* 2010, 2011) and many other natural and anthropogenic phenomenon (e.g. Fernandez *et al.* 2009; Gonzalez & Fernandez 2011). Manzo *et al.* (2006) proposed methodology for calculating east–west and vertical components of mean deformation rates when both ascending and descending InSAR data with overlapping temporal coverage are available. Wright *et al.* (2004) suggested that having ascending and descending right and left looking InSAR data is sufficient for reconstructing 3-D fault displacement. Integration of ascending and descending data from the same satellite was recently presented in Ozawa & Ueda (2011).

In this paper, we present a methodology that allows direct integration of various InSAR data sets for producing 2-D time-series of deformation when two or more InSAR data sets with overlapping temporal and spatial coverage are available. The technique is based on the SBAS method that is modified to account for variation in sensor parameters. It allows integration of data with different

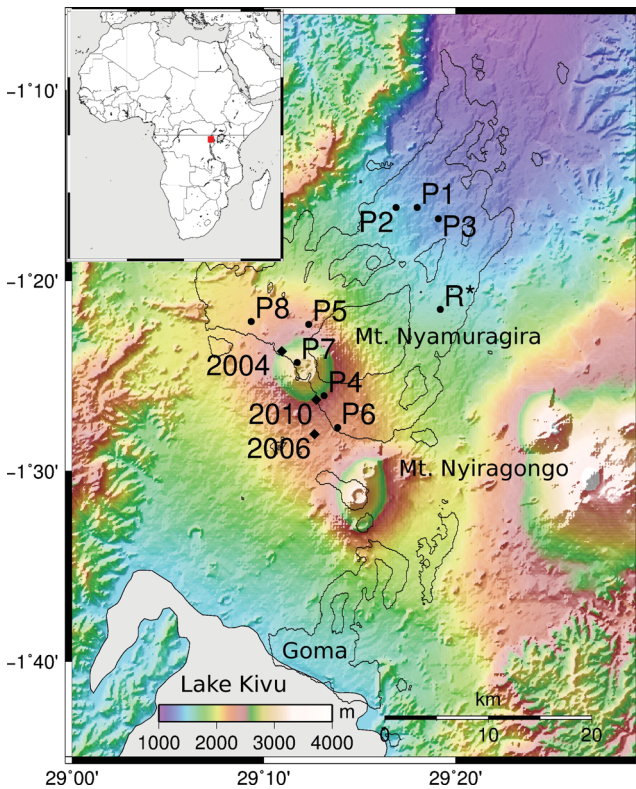


Figure 1. Virunga Volcanic Province. Background image is 30 m resolution SRTM DEM. Coherent area common to all images acquired during 2006–2010 is outlined in black. 2004, 2006 and 2010 eruptive centres are shown as black diamonds. P1–P8 regions are studied in detail in the paper. City of Goma, Lake Kivu, Mt Nyamuragira and Mt Nyiragongo are shown.

wave-band, azimuth and incidence angles, different spatial and temporal sampling and resolution, including air-borne and space-borne data from sensors with varying parameters. In addition to presenting geophysical results we demonstrate results of numerical simulation and describe methods for smoothing produced multidimensional time-series. Such analysis is important for understanding the accuracy and limitations of the proposed methodology. We do not intend to model observed ground deformation but we want to provide here only qualitative analysis.

2 GEOLOGICAL CONTEXT

We apply this technique to studying volcanic deformation of the Virunga Volcanic Province (VVP), North Kivu, Democratic Republic of Congo located along the Western branch of the East Africa Rift (EAR; Fig. 1). The EAR is a major tectonic feature that shapes Central Africa and defines the Rift Valley, a lowland area between highland ranges which formed due to the action of geological faults and is associated with earthquakes and volcanoes. The VVP is a transfer zone between two segments of the Western Branch of the EAR (Chorowicz 2005). The VVP marks the northern end of the Kivu basin, a 100 km-long and 30 km-wide half-graben hosting Lake Kivu. The VVP constitutes the transition between the Kivu basin to the South and the Lake Edward basin to the North (Ebinger 1989).

The VVP has been active since the mid-Miocene, its active volcanoes were however only discovered at the end of the 19th century. Currently, the VVP experiences low seismicity but hosts, among its eight main volcanic edifices, two of the most active volcanoes of

Africa: Nyamuragira and Nyiragongo. The first erupted about 30 times over the last century (Smets *et al.* 2010) whereas the second hosts in its crater what is currently believed to be the largest (semi-) permanent lava lake on Earth.

During the two sole historical eruptions known since its discovery (i.e. 1977 and 2002), the Nyiragongo lava lake was drained through a network of fractures that opened from the volcano up to the nearby city of Goma located 15 km to the south (Tazieff 1977; Durieux 2004; Komorowski *et al.* 2004; Tedesco *et al.* 2007). The highly silica-undersaturated lavas flowed at high speed and destroyed entire villages and a part of the city of Goma within a few hours.

In contrast, Nyamuragira lava flows affect only the equatorial forest of the Virunga National Park, although when flowing towards the south, lava can reach inhabited areas and even the shore of Lake Kivu as it happened in 1938 and 1948 (Smets *et al.* 2010).

On 2010 January 2 a 600 m-long fracture opened along the southern flank of the volcano after less than an hour of precursory seismic activity (BGVN 2010) and a 10 km-long lava flow devastated more than 900 ha of forest in less than 3 weeks. The previous eruption occurred almost at the same place on 2006 November 27, but it was preceded by more than 1.5 days of precursory long-period seismic swarms.

These Nyiragongo and Nyamuragira lava flows cutting through the equatorial vegetation constitute large zones where the satellite radar signals remain coherent over time. This offers the possibility of using satellite radar interferometry (InSAR) to study the ground deformation associated with these eruptions (d'Oreye *et al.* 2008; Wauthier *et al.* 2012).

In this study, we make use of eight independent data sets: six ENVISAT, one ALOS PALSAR and one RADARSAT-2 fine beam set spanning all together about eight years from 2003 to 2010 in both ascending and descending geometries (Table 1, Fig. 2). We perform two runs. The first run utilizes only three tracks from ENVISAT spanning 2006–2010. This run has shorter temporal span but better spatial coverage due to larger common footprint of the three data sets. The horizontal and vertical time-series produce by the second run based on eight data sets spanning 2003–2010 were calculated from over a thousand interferograms.

3 METHODOLOGY

In case of a single set of SAR acquired by a sensor with an azimuth θ and an incidence angle ϕ the time-series of deformation can be reconstructed by applying the SBAS method (Berardino *et al.* 2002; Usai 2003)

$$AV_{\text{los}} = \Phi_{\text{obs}}, \quad V_{\text{los}} = A^+ \Phi_{\text{obs}}, \quad d_{\text{los}}^{i+1} = d_{\text{los}}^i + V_{\text{los}}^{i+1} \Delta t^{i+1}, \quad (1)$$

where A is a matrix constructed from the time intervals between consecutive SAR acquisitions, V_{los} is a vector of the unknown line-of-sight velocities, Φ_{obs} is a vector of observed interferogram values, A^+ is a pseudo-inverse of matrix A found by applying the singular value decomposition (SVD), and d_{los}^i is a line-of-sight displacement at the time t^i . The problem stated by the eq. (1) is over-determined when the number of linear independent equations M^* is equal to the number of unknown velocities: $M^* = (N - 1)$, where N is a number of SAR images used in processing, and the total number of equations (i.e. observed interferograms) is greater than the number of unknown velocities: $M > (N - 1)$. The problem is under-determined when the number of linear independent equations is less than the number of unknown velocities: $M^* < (N - 1)$.

Table 1. SAR data sets used in this work: time span (in YYYYMMDD format), azimuth θ and incidence ϕ angles, number of available SAR images N and number of calculated interferograms M .

InSAR set	Time span	θ (°)	ϕ (°)	N	M
ENVISAT, Track 035IS2 (dsc)	20030116–20100916	−168	25	42	224
ENVISAT, Track 450IS7 (dsc)	20060519–20100910	−168	44	30	169
ENVISAT, Track 314IS7 (asc)	20060613–20100831	−12	44	41	308
ENVISAT, Track 228IS2 (asc)	20021225–20061025	−12	23	33	53
ENVISAT, Track 042IS5 (asc)	20080424–20100916	−12	38	20	96
ENVISAT, Track 493IS4 (dsc)	20080421–20100913	−168	34	18	86
ALOS, Track 580 (asc)	20071027–20100504	−12	39	9	36
RADARSAT-2, F21 (dsc)	20091215–20110527	−168	35	16	79
Total (only used images):	20030116–20100916			181	1051

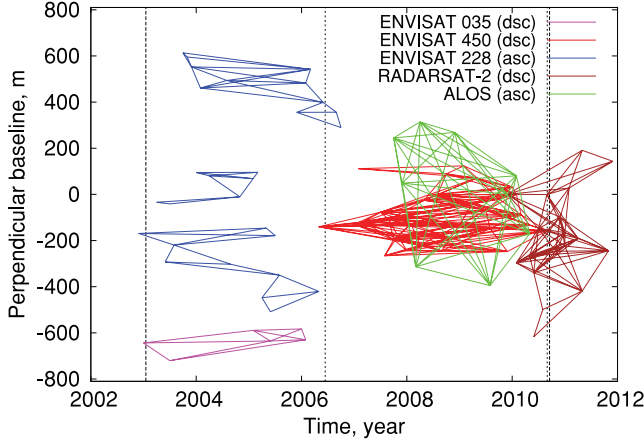


Figure 2. Distribution of interferograms and their perpendicular baselines for selected data sets used in these study. For ENVISAT track 035 (dsc), we plotted only interferograms acquired before 2007 using baselines corresponding to actual master image acquired on 2010 January 14 (not shown here). Additional data sets not shown here mostly span 2006–2010 time period. Vertical lines correspond to start and end time of investigated periods, internal lines correspond to first (with three InSAR sets) run and external lines correspond to second (with eight InSAR sets) run.

In case of K multiple SAR sets acquired by sensors with different orbital parameters (e.g. azimuth and incidence angles) the eq. (1) can be rewritten in the following form for each set $k = 1 \dots K$

$$|S_N^k A \quad S_E^k A \quad S_U^k A| \cdot |V_N \quad V_E \quad V_U|^T = \Phi_{\text{obs}}^k \quad (2)$$

assuming that $V_{\text{los}} = \mathbf{S}\mathbf{V} = S_N V_N + S_E V_E + S_U V_U$ and $\mathbf{S} = \{S_N, S_E, S_U\} = \{\sin \theta \sin \phi, -\cos \theta \sin \phi, \cos \phi\}$, where $\mathbf{S}$ is a line-of-sight unit vector with north, east and up components S_N, S_E, S_U and $\mathbf{V}$ is a velocity (ground deformation rate) vector with components V_N, V_E, V_U .

Then a multidimensional SBAS (MSBAS) method that includes all K sets of independently acquired SAR data can be presented in the following form:

$$\begin{pmatrix} A^1 \\ A^2 \\ \dots \\ A^K \end{pmatrix} \begin{pmatrix} V_N \\ V_E \\ V_U \end{pmatrix} = \begin{pmatrix} \Phi^1 \\ \Phi^2 \\ \dots \\ \Phi^K \end{pmatrix} \quad \text{or} \quad \hat{A} \hat{V}_{\text{los}} = \hat{\Phi}_{\text{obs}}, \quad (3)$$

where the new matrix $\hat{A}$ (as in 2) has dimensions $3(\sum_{k=1}^K N^k - 1) \times \sum_{k=1}^K M^k$, the new vector $\hat{V}$ has dimensions $1 \times 3(\sum_{k=1}^K N^k - 1)$, and the new vector $\hat{\Phi}_{\text{obs}}$ has dimensions $1 \times \sum_{k=1}^K M^k$.

This problem is usually under-determined because the number of linearly independent equation is less than the number of unknown

velocities. Indeed, if all SAR images were acquired at different times t^i then the number of unknowns would be equal to $3(\sum_{k=1}^K N^k - 1)$ and the maximum possible number of independently observed interferograms would be equal to $(\sum_{k=1}^K N^k - K)$. Additional constraints that increase rank of $\hat{A}$ are introduced by utilizing variability in directional cosines S_N, S_E, S_U applied to rows of this matrix.

All modern space-borne SAR systems orbit the earth in a near-polar orbit and can acquire data only in two independent acquisition geometries: ascending and descending. Such acquisition geometries are not very sensitive to a motion in northern direction (i.e. along track). Therefore, the number of unknowns in the eq. (3) can be reduced to $2(\sum_{k=1}^K N^k - 1)$ by excluding all terms responsible for northern motion V_N . Such approximation is reasonable when the magnitude of north–south component of deformation is comparable to (not significantly larger than) the magnitude of east-west and vertical components (Wright *et al.* 2004).

Solution of the problem stated by the eq. (3) is found by applying SVD to $\hat{A}$. In a least square sense an unlimited number of solutions of (3) exists but only minimum norm solution is selected by SVD. Due to rank deficiency of the stated problem the calculated solution oscillates around the true but unknown solution. In the following sections of this paper, we refer to such solution as a raw solution.

To remove oscillations caused by a rank deficiency of the problem (3) we apply Tikhonov regularization (Tikhonov & Arsenin 1977). The regularized problem can be written in the following form:

$$\begin{pmatrix} \hat{A} \\ \lambda I \end{pmatrix} \begin{pmatrix} V_E \\ V_U \end{pmatrix} = \begin{pmatrix} \hat{\Phi} \\ 0 \end{pmatrix}, \quad (4)$$

where λ is a regularization parameter that can be found, for example, using L-curve method (Hansen & O’Leary 1993) and I is a $2(\sum_{k=1}^K N^k - 1) \times 2(\sum_{k=1}^K N^k - 1)$ identity matrix. Tikhonov regularization is one of the possible methods for regularization of ill-posed problems, which solutions either do not exist, non-unique or unstable. The L-curve method is based on the plot for all regularization parameters of the size of the regularized solution versus the corresponding residual. Using this method the optimal parameter λ is selected at the intersection of the vertical and horizontal lines of the ‘L’. The practical example of Tikhonov regularization with software code can be found in section 19.5 (Linear Regularization Methods) of Press *et al.* (2007). Such a solution is furthermore called a ‘regular solution’.

An alternative to regularization is low-pass filtering in the time domain. In this case we apply Gaussian smoothing (e.g. Gonzalez & Woods 2001) that does not require an uniform temporal spacing. Such solution is called a filtered solution from here on.

From a computational point of view the regularization approach is preferable since it is applied during the inversion of matrix $\hat{A}$ and

does not require any additional time. On the other hand, the filtering approach is extremely time consuming; depending on the number of images and the extent of the area the filtering procedure may require several hours to complete.

3.1 Theoretical considerations

The problem (3) is rank deficient when acquisition times from different SAR data sets do not precisely match. One possible solution proposed in Usai (2003) is to recreate missing data by interpolation. In this case problem (3) becomes well conditioned and can be solved precisely without need for regularization. However, such solution in general is of a low precision because of atmospheric, orbital, thermal and interpolation noise presented in original and recreated by the interpolation data. Such approach is undesirable because of ambiguity in type of interpolation and selection of points at which interpolation is performed and also because of computational limitations. Lanari *et al.* (2004) showed that in case of sufficiently dense coverage rank deficiency does not introduce significant degradation of precision. Numerical simulations presented in the following section of this paper suggest that even significant rank deficiency does not degraded precision in case of sufficiently dense coverage. From practical examples presented in Lanari *et al.* (2004) and in this paper and from theoretical considerations it follows that it is not rank deficiency that is important but rank itself. Indeed, if rank of a problem is equal to 187 (as in our case) then it is possible to assume that a model can be precisely described by 187 parameters. These parameters will be defined with high precision because they will be calculated from more than 187 original observations without any data manipulation. Therefore, instead of fitting data using (3) this problem can be reformulated as a parameter estimation problem as following:

$$\hat{A}BX = \hat{\Phi}_{\text{obs}}, \quad (5)$$

where X is a set of unknown model parameters with dimension equal to rank of matrix $\hat{A}$ and $\hat{A}B$ is a matrix describing a chosen model.

Assuming the parameter estimation approach, it is apparent that availability of 187 parameters based on observations more or less uniformly distributed over the entire observation period is more than enough for defining even very complicated model (e.g. high degree polynomial). In our follow-up work we will provide examples of model parameter estimation but in this paper we assume data fitting approach and rely on regularization or filtering for reducing effects of rank deficiency.

3.2 Numerical simulations

For the numerical simulations we used real matrix $\hat{A}$ calculated from eight InSAR data sets (Table 1, Fig. 2) that will be described in more detail in the next section. This matrix consists of 360 (or $2(\sum_{k=1}^K N^k - 1)$) columns corresponding to 181 SAR images and 1051 rows corresponding to 1051 interferograms (overlapping in time, ascending and descending). Rank of this matrix calculated with the Linear Algebra PACKage (LAPACK <http://www.netlib.org/lapack>) is equal to 187.

We created 1051 synthetic interferograms assuming the following model (somewhat resembling ground deformation signal observed in VVP). We assumed synthetic ground motion with constant velocity rate of 2 cm yr^{-1} in east–west direction and -4 cm yr^{-1} in vertical direction. Additional periodic signal with amplitude of 2 cm and period of 6 months was added to east–west component.

Large co-eruptive displacement with magnitude of 12 cm occurred in 2009 was added to the east–west component and two reversing co-eruptive displacements with magnitude of $\pm 20 \text{ cm}$ occurred in 2007 and 2009 were added to the vertical component. The first run was performed with uncontaminated by the noise data, and for the second and third runs we added to the synthetic interferograms Gaussian noise with standard deviation 1.4 and 2.8 cm (approximately 0.5 and 1 C-band fringe). Raw time-series were reconstructed by solving the problem stated by the eq. (3) and regularized time-series were reconstructed by solving the problem stated by the eq. (4) assuming $\lambda = 0.25$. Filtered time-series were produced using Gaussian smoothing with window equal to 1 month applied to raw time-series.

Results of simulations for interferograms with added Gaussian noise with standard deviation of 1.4 cm are presented in Figs 3(a)–(c), and with standard deviation of 2.8 cm are presented in Figs 3(d)–(f). In these images the output was averaged between eleven neighbouring pixels to reduce the effect of added noise. By varying regularization parameter λ and the width of Gaussian window during filtering, the magnitude of smoothness can be manipulated. All runs were computed independently, therefore, noise distribution in individual interferograms in Fig. 3 varied.

It can be seen that both east–west and vertical time-series are reconstructed with remarkable precision, and as expected, precision of vertical component is better than precision of east–west component. It also can be seen that precision during 2003–2007 is somewhat lower than precision during 2007–2010. This effect can be explained by the difference in temporal resolution and number of interferograms used.

To provide quantitative analysis of precision we calculated various statistical parameters presented in Table 2. Here we estimated the root mean square (rms) error between synthetic and reconstructed time-series. These values range approximately 0.5–1.5 cm depending on the magnitude of the added noise and processing level. The best precision 0.42 cm was achieved by the vertical component of raw time-series in absence of noise and the worst 1.52 cm precision was achieved by the east–west component of regularized time-series also in absence of noise. The correlation coefficient between synthetic and reconstructed time-series ranged 0.98–0.99 and is not shown in Table 2.

Reason for lower precision of the regularized and filtered time-series is over-smoothing of periodic and step signals. This undesirable effect is common to all regularization and filtering procedures and cannot be entirely avoided but its magnitude can be minimized by weakening the strength of the regularization and filtering parameters.

To validate capability of this technique for detecting pre-eruptive and pre-seismic signals we estimated the rms error and correlation coefficient between synthetic and reconstructed velocities produced by the MSBAS processing. We believe that the statistical parameters of velocities are more informative than those of reconstructed displacements because the latter are non-stationary. In Table 2 it can be seen that velocities are reconstructed by this method with the rms close to 1 cm yr^{-1} for east–west component and $0.4\text{--}1 \text{ cm yr}^{-1}$ for vertical component. Correlation coefficients for all runs are larger than 0.5 and often are larger than 0.75. Such high correlation suggests that displacements reconstructed from calculated velocities are likely to be of the correct sign as the true displacements even that their magnitude is estimated with somewhat moderate precision. Qualitatively, it can be concluded that availability of a few measurements obtained consequently and showing displacements of the same sign with magnitude above 1 cm is sufficient to

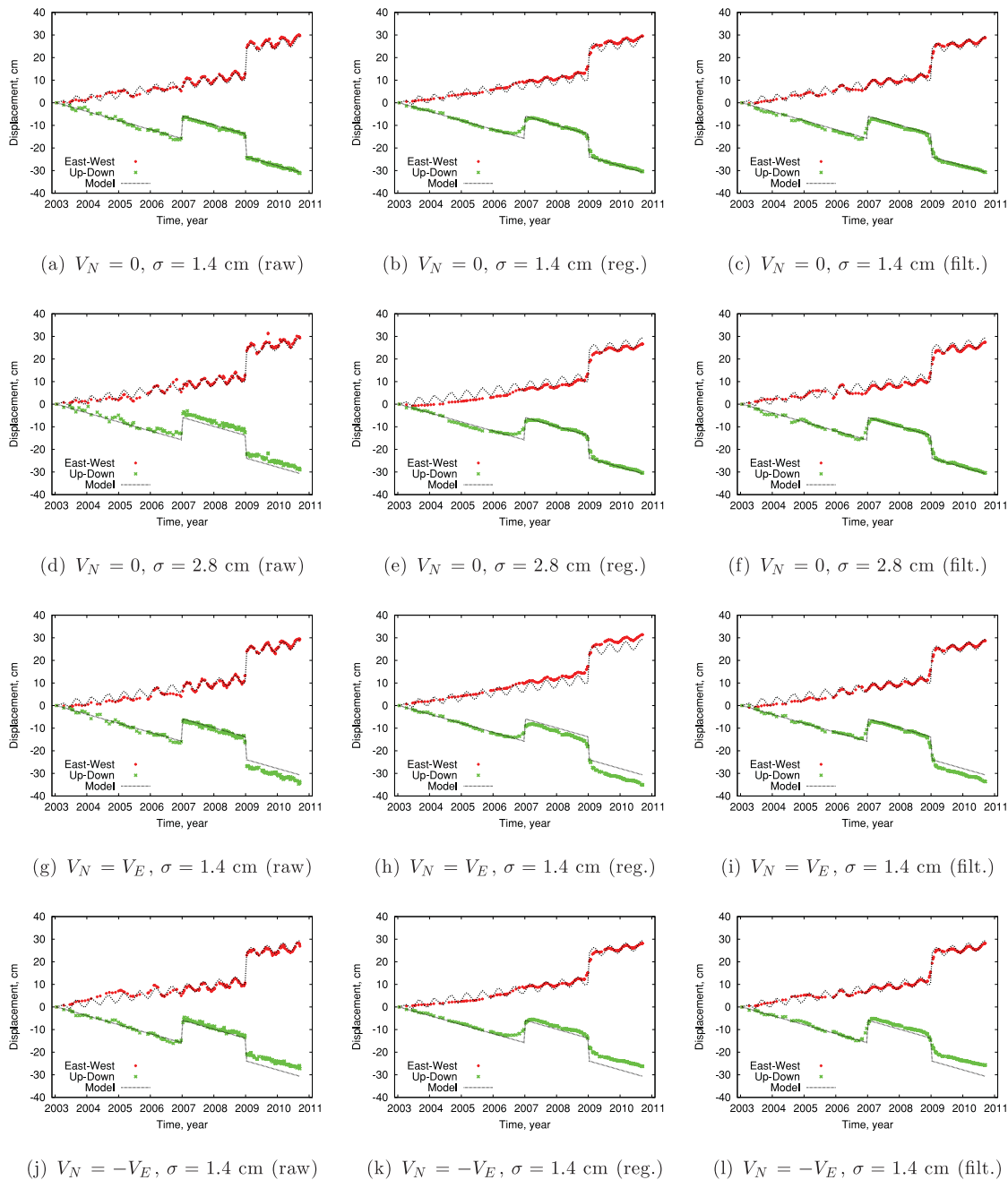


Figure 3. Results of numerical simulations. Modelled synthetic ground motion (black dots) and reconstructed time-series in east–west (red) and vertical (green) directions recovered by proposed method. Raw (first column), regularized (second column) and filtered (third column) time-series are shown.

interpret as a reliable deformation signal. Quantitative estimation of reliability of this technique for mapping pre-event deformation depends on the number, precision and distribution of the particular measurements and needs to be studied further.

We also estimated precision of our technique in case of large north–south motion. For this we produced two sets of synthetic time-series assuming presence of north–south motion of the same magnitude as east–west motion and of the same and oppose signs ($V_N = V_E$ and $V_N = -V_E$). These results are shown in Figs 3(g)–(i) and 3(j)–(l) and statistical parameters are shown in the second and third parts of Table 2.

Qualitatively precision of reconstructed time-series is remarkable. The precision of vertical component of displacements de-

creased approximately twice to 1.4–1.9 cm and precision of east–west component practically did not change. A somewhat larger discrepancy between synthetic and observed vertical time-series is observed after a large step displacement occurred in 2009. Precision and correlation coefficient of velocities did not change significantly. It is apparent that reconstructed time-series clearly describe ongoing deformation pattern and can be used for modelling without further modification. Nevertheless, further work needs to be performed to better understand pattern of observed discrepancies. Such work is outside of the scope of this paper as it is not directly related to MSBAS methodology proposed here but is due to the effect of 2-D approximation of 3-D motion. It is believed that similar effect will be observed by other techniques (e.g. Manzo *et al.* 2006).

Table 2. Statistical analysis of numerical simulation results. Root mean square error (rms) between modelled and reconstructed displacements and velocities. Correlation coefficients between modelled and reconstructed displacements were estimated (not shown here) that ranged 0.98–0.99. Correlation coefficients between modelled and reconstructed velocities for east–west and vertical components (γ_{EW} and γ_{Up}) are shown in last two columns. Increase in rms for regularized and filtered time-series is due to over-smoothing periodic and step signals.

Processing	σ (cm)	rms EW (cm)	rms Up (cm)	rms EW (cm yr ⁻¹)	rms Up (cm yr ⁻¹)	γ_{EW}	γ_{Up}
Raw, $V_n = 0$	0	0.74	0.42	0.68	0.41	0.86	0.92
Raw, $V_n = 0$	1.4	0.86	0.48	0.76	0.47	0.82	0.90
Raw, $V_n = 0$	2.8	1.13	0.87	0.99	0.79	0.71	0.78
Reg., $V_n = 0$	0	1.52	0.80	1.03	0.70	0.73	0.82
Reg., $V_n = 0$	1.4	1.50	0.78	1.03	0.71	0.74	0.81
Reg., $V_n = 0$	2.8	1.46	0.90	1.02	0.72	0.75	0.79
Filt., $V_n = 0$	0	1.24	0.83	1.03	0.88	0.73	0.57
Filt., $V_n = 0$	1.4	1.31	0.84	1.06	0.89	0.69	0.55
Filt., $V_n = 0$	2.8	1.35	0.89	0.99	0.92	0.71	0.51
Raw, $V_n = V_e$	0	1.06	1.39	0.76	0.61	0.82	0.86
Raw, $V_n = V_e$	1.4	1.04	1.35	0.76	0.69	0.82	0.83
Raw, $V_n = V_e$	2.8	1.75	1.38	0.86	0.79	0.77	0.79
Reg., $V_n = V_e$	0	1.57	1.74	1.02	0.68	0.74	0.82
Reg., $V_n = V_e$	1.4	1.65	1.74	1.02	0.69	0.74	0.81
Reg., $V_n = V_e$	2.8	1.84	1.61	1.02	0.68	0.73	0.81
Filt., $V_n = V_e$	0	1.39	1.46	1.01	0.87	0.74	0.58
Filt., $V_n = V_e$	1.4	1.29	1.40	1.00	0.88	0.73	0.57
Filt., $V_n = V_e$	2.8	1.34	1.51	1.05	0.87	0.66	0.59
Raw, $V_n = -V_e$	0	0.80	1.70	0.73	0.56	0.84	0.85
Raw, $V_n = -V_e$	1.4	1.20	1.64	0.79	0.61	0.81	0.83
Raw, $V_n = -V_e$	2.8	1.45	1.67	0.88	0.85	0.76	0.70
Reg., $V_n = -V_e$	0	1.50	1.66	1.05	0.74	0.72	0.81
Reg., $V_n = -V_e$	1.4	1.52	1.68	1.05	0.74	0.71	0.81
Reg., $V_n = -V_e$	2.8	1.74	1.65	1.07	0.75	0.67	0.78
Filt., $V_n = -V_e$	0	1.30	1.85	1.05	0.90	0.71	0.54
Filt., $V_n = -V_e$	1.4	1.27	1.84	1.03	0.91	0.74	0.53
Filt., $V_n = -V_e$	2.8	1.57	1.91	1.19	0.94	0.45	0.47

It is worth emphasizing that this MSBAS technique similar to the standard SBAS technique computes velocities (ground deformation rates) between consecutive acquisitions and time-series of ground deformation are reconstructed by integration. Therefore, if any velocity value at any particular time is calculated with an error (due to atmospheric noise, large step displacement, rank deficiency, or 2-D approximation of 3-D motion) the following displacements will be offset by the value of the error. In case of complete rank and good coverage this offset will converge to zero with time. In case of rank deficient problem this offset will most likely remain uncompensated, and will need to be manually corrected during validation. This effect is observed, for example, in vertical component in Figs 3(g)–(i) and 3(j)–(l) after 2009. Fortunately, this effect is not largely dependent on number of data sets and good results are achieved even with only one ascending and descending sets.

4 APPLICATION TO VIRUNGA VOLCANIC PROVINCE

The proposed methodology was used for calculation of the ground deformation time-series in the VVP, Democratic Republic of the Congo (Fig. 1). The 2003–2010 time-series spanned three volcanic eruptions at Mt Nyamuragira occurred on 2004 May 8, 2006 November 27 and 2010 January 2.

For this we collected eight independent SAR data sets (Table 1): six ENVISAT (three ascending and three descending),

one ALOS PALSAR (ascending) and one RADARSAT-2 (descending). Each data set was processed independently with GAMMA software (Wegmuller & Werner 1997). Single master was selected for each data set and all single look complex (SLC) images were co-registered to the selected master. All possible interferograms with perpendicular baselines less than 100 m for ENVISAT, 200 m for RADARSAT-2 and 700 m for ALOS were created. The topographic phase was removed using 30 m SRTM DEM (Farr & Kobrick 2000). Differential interferograms were filtered, unwrapped and geocoded on the 30 m SRTM DEM grid. Before geocoding an orbital correction algorithm was applied to remove linear ramps in interferograms. The entire processing was performed automatically and only one ALOS PALSAR image (acquired on 2009 September 16) contaminated by significant atmospheric noise was manually removed.

Two runs were performed. The first run utilized only three ENVISAT tracks (035IS2, 450IS7, 314IS7, first three data sets in Table 1) that have better spatial but a shorter (2006–2010) temporal coverage. For this run linear deformation rates for east–west and vertical components calculated from regularized time-series are shown in Fig. 4 and corresponding time-series for seven selected regions (11 × 11 pixels, points P1–P8, excluding P5) are plotted in Figs 5 and 6. Deformation signals are clearly observed in raw, regularized and filtered time-series.

The second run utilized all eight InSAR data sets and consisted of 181 SAR images and 1051 interferograms. The temporal coverage

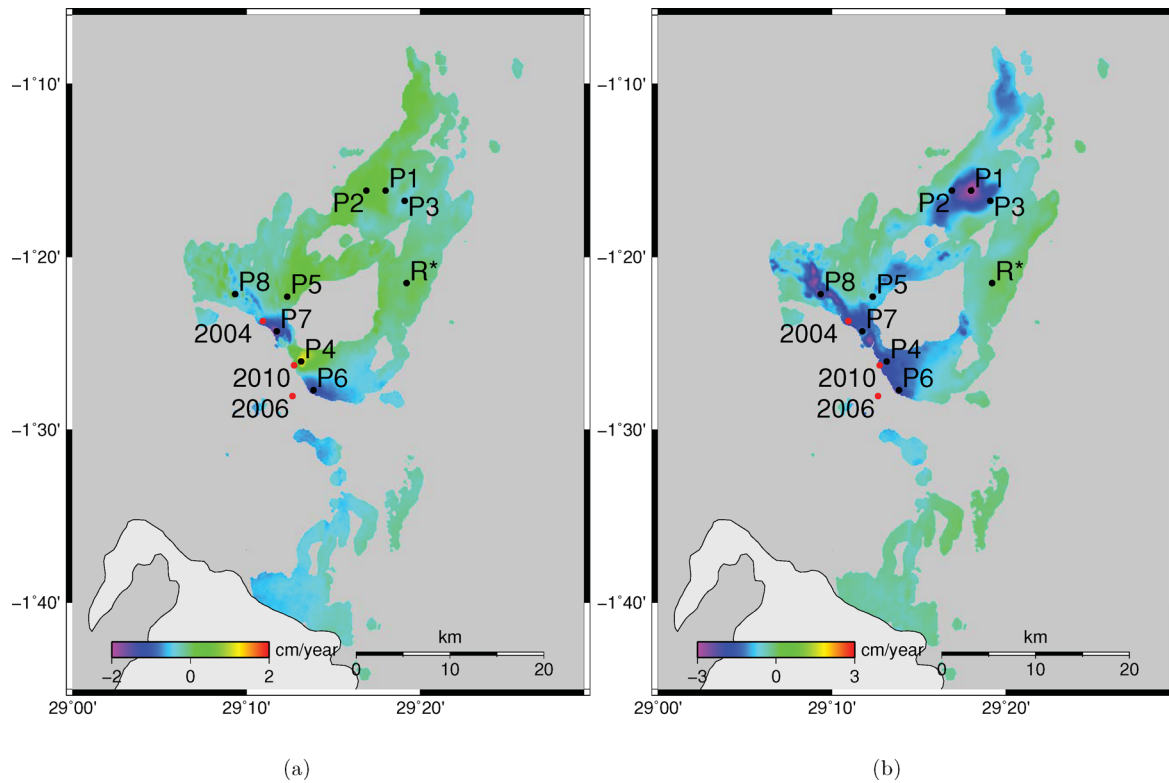


Figure 4. 2006–2010 east–west (left-hand panel) and vertical (right-hand panel) linear deformation rates calculated from first three data sets shown in Table 1. For points P1–P8 (excluding P5) we present time-series below. Region R* was used as stable reference during calculation of 2006–2010 linear deformation rates. 2004, 2006 and 2010 eruptive centres are shown as red points.

of this run is longer, from 2003 to 2010, but the spatial coverage is reduced. The common footprint of the eight data set does not include the largest subsidence region marked by points P1–P3. Spatial coverage of this run is also reduced along a small area NW of Nyamuragira, where the 2004 lava flow were emplaced atop of the 1998 lava (Smets *et al.* 2010). Linear deformation rates and corresponding time-series for points P4–P8 are shown in Figs 7 and 8. It is apparent that both sets of time-series are in a very good agreement in overlapping time period. This suggests that the solution does not significantly depend on the number of SAR images or a selection of reference region. However, the improved temporal resolution provided by the second run allows reconstruction of the deformation signal in more detail.

In Table 3 we present the rms and correlation coefficient between time-series of both runs for common points P4, P6, P7 and P8 during 2006–2010. The rms error ranges from 0.5 to 1.1 cm and is smallest for regularized time-series. Vertical component consistently has smaller error than east–west component. Correlation coefficient is in range 0.87–0.99 with exception of one point P8 with negligibly small east–west motion. For this point P8 for east–west component the rms is about 0.7–1 cm but correlation coefficient is close to zero. The rms error calculated for these time-series is in a very good agreement with the rms calculated for the synthetic time-series.

4.1 Geophysical description

Vertical and east–west time-series produced by the proposed technique are used for analysing ground deformation occurred in the VVP during 2003–2010. By definition positive displacements are

eastward and upward and negative displacements are westward and downward.

The lava field north of Nyamuragira is observed only on the 2006–2010 time-series due to the superior spatial coverage common to three data sets. Fig. 5 shows vertical and east–west time-series for points P1, P2 and P3. These points are located in the centre of the large lava flow field located N-NE of Nyamuragira volcano (Fig. 4), where multiple lava flows piled up over recent and historic time (1991–93, 1980, 1967, 1958 and others). This area is known for being affected by the monotonous subsidence attributed to lava compaction and relaxation of the substrate (Colclough 2005). This phenomenon is commonly observed on other volcanoes (e.g. Amelung *et al.* 2000; Stevens *et al.* 2001; Lu *et al.* 2005; Peltier *et al.* 2010).

Point P1, located at the centre of the lava flow field, is affected by the fastest, smooth downward movement of about -4 cm yr^{-1} , with no horizontal component. Similarly points P2 and P3, located correspondingly west and east of P1, are affected by the smooth though smaller linear downward movement of about -1.5 cm yr^{-1} . Unlike P1, they are also affected by a horizontal movement of correspondingly 0.5 cm yr^{-1} and -0.5 cm yr^{-1} towards the centre of the subsiding area. Although we have no information about the exact amount and thickness of the flows emplaced there, it is reasonable to assumed that they occupied a depression, resulting in a thicker pile at the centre where the highest rate of subsidence is observed and towards which inward displacements are expected (Samieie-Esfahany *et al.* 2009).

Displacements near and within the Nyamuragira main crater can be studied using both the 2006–2010 and 2003–2010 data sets (Figs 4 and 7). Results of vertical and east–west time-series are consistent between the two sets (Figs 6 and 8) with the exception

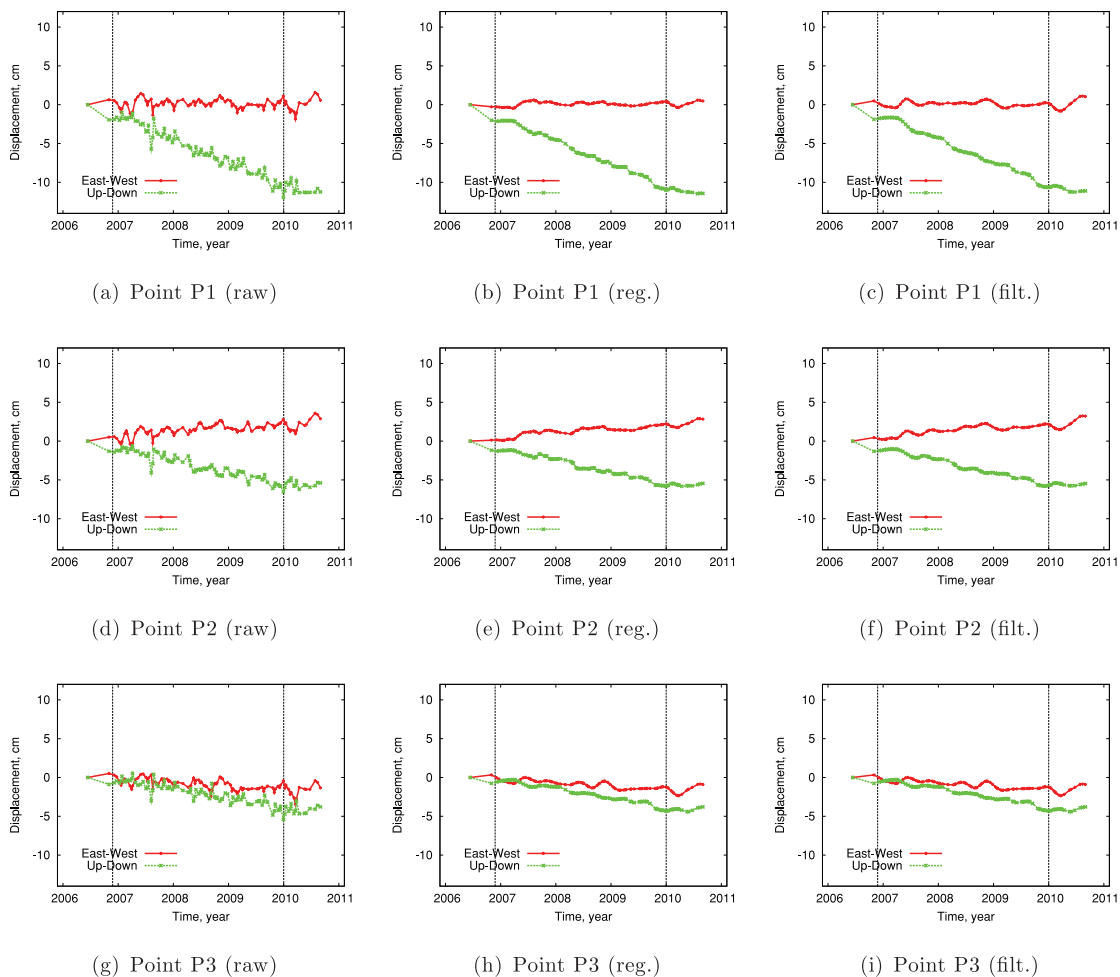


Figure 5. 2006–2010 east–west and vertical time-series of ground deformation at points P1 (top row), P2 (middle row) and P3 (bottom row). Raw (first column), regularized (second column) and filtered (third column) time-series are shown. Vertical bars mark 2006 November and 2010 January eruptions.

of a change of the subsidence rate observed after the 2010 January eruption. We believe that this discrepancy is due to motion of the reference point chosen for the 2006–2010 data set.

Point P4 and P6 are located on the SE flank of Nyamuragira, respectively close to the 2010 and 2006 eruptive fractures. Point P5 is located on the N-NE flank, point P7 is located close to the NW rim of the Nyamuragira main crater and point P8 is located NW of Nyamuragira (Fig. 7). Unambiguous changes in long-term vertical and horizontal movements are clearly defined and co-eruptive displacements associated with each eruption are clearly visible on both horizontal and vertical components (Fig. 8).

Pre-eruptive signals are detected for the 2010 eruption about 15 days prior the onset of the eruption on images acquired on December 15, 17 and 29 (see star and arrow in Fig. 9(c) that marks beginning of the pre-eruptive deformation), whereas the seismic precursors only started less than 1 hr before the lava outburst (BGVN 2010). The pre-eruptive deformation signals are of about the same amplitude and spatial extent as the atmospheric noise and, therefore, cannot be identified on individual differential interferograms. Conventional single geometry SBAS method, as it typically combines images acquired every 24–35 days (RADARSAT and ENVISAT respectively) does not benefit from the sufficient time sampling to detect an outlier and confirm it with the second acquisition before the occurrence of the eruption. Because of high temporal sampling, this is the first time that pre-eruptive geodetic signals can unambiguously be de-

tected for a volcano in the VVP. Similar pre-eruptive signals are also expected for the previous eruptions but, as the dense systematic InSAR monitoring of the area only started at the end of 2006 (d'Oreye *et al.* 2008; d'Oreye & Celli 2010), the amount of available images is insufficient for accurate detection. It is worth emphasizing that spatial averaging used to plot the time-series (11×11 pixels) may significantly reduce the estimated amplitude of localized steep deformation such as co-eruptive deformation shown here. Nevertheless, our results confirm pre-eruptive signals suggested by Wauthier (2011). Using conventional time-series methods based on a single acquisition geometry she observed a sharp range change attributed to pre-eruptive movements. The amplitude of those displacements was however of the same order of magnitude as noise. The absence of other images acquired before the eruption did not allow them to discriminate the observed signal from a possible atmospheric artefact.

In the next paragraphs we show examples of some of the most striking features detected in the vertical and horizontal time-series associated to these three eruptions. Their analysis and interpretation are however beyond the scope of this paper.

The 2004 May 8–28 eruption

Subsidence of about -2 – 3 cm and horizontal eastward displacement of 5 – 6 cm associated with the 2004 eruption are detected at point

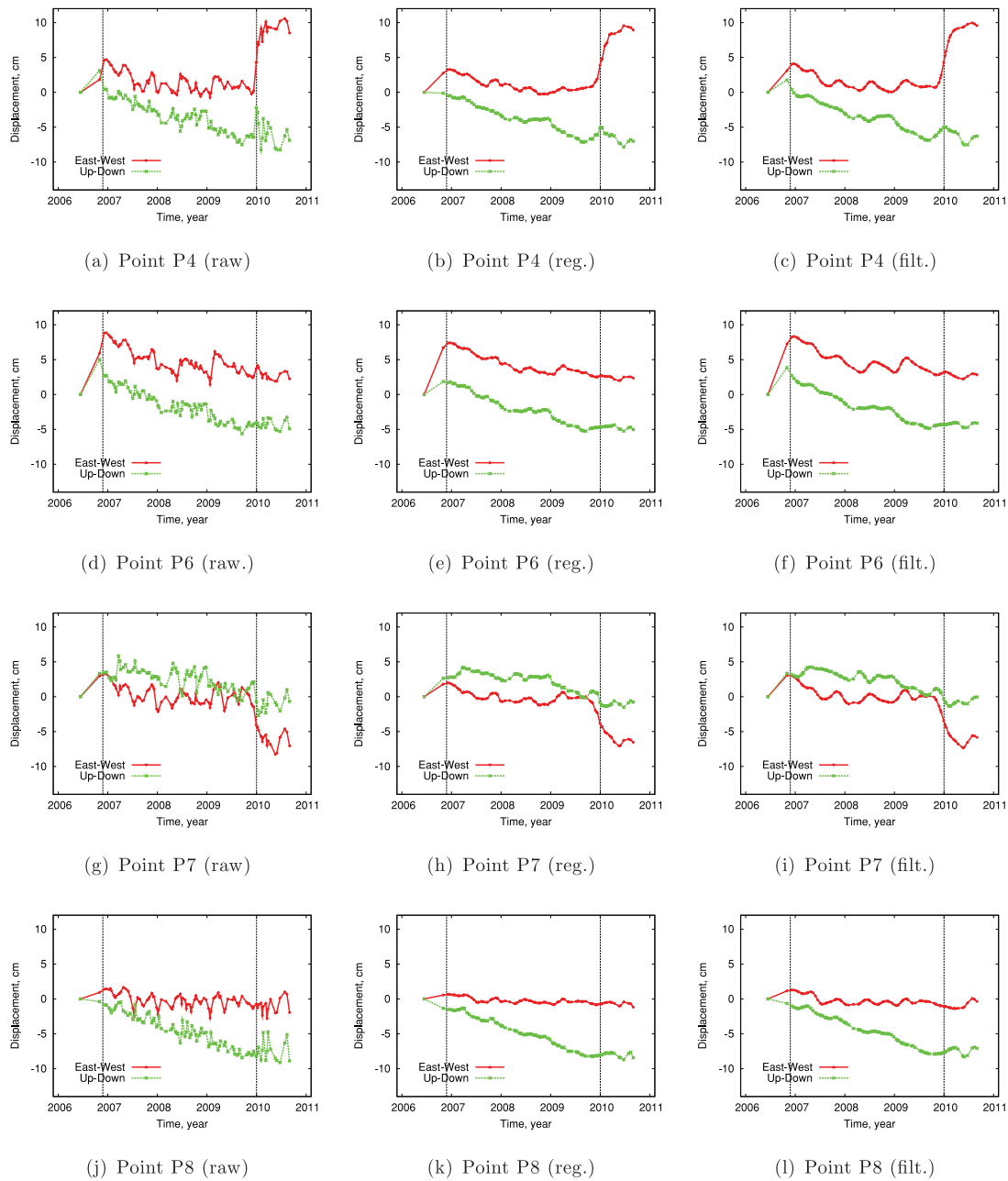


Figure 6. 2006–2010 east–west and vertical time-series of ground deformation at points P4, P6, P7 and P8 (rows top to bottom). Raw (first column), regularized (second column) and filtered (third column) time-series are shown.

P5 located on the N–NW flank of the volcano (Figs 8d–f and 9a). About one third of that horizontal eastward movement (1–2 cm) seems to have started around mid-February 2004 (see star and arrow in Fig. 9a), which is consistent with the increase of long period seismicity recorded 2–4 months before the eruption (Mavonga *et al.* 2006).

A 2–3 cm horizontal eastward displacement is also detected at point P7 on the NW rim of the crater (Figs 8j–l and 9d). About half of that horizontal eastward movement took place at least 10 days before the eruption (see star and arrow in Fig. 9d). Note that the eruption started with volcanic activity within the caldera before propagating to an eruptive fracture along the N flank (BGVN 2004). From 2005, horizontal movement reverted and that portion of the crater started to move westward at a rate of about -1.5 cm yr^{-1} , a drift that stopped at the end of 2008.

The 2006 November 27–December 5 eruption

Point P6, located along the SE flank of Nyamuragira, did not experience significant horizontal motion until a -2 cm westward displacement occurred at the end of 2005 (Fig. 8g). Since early 2006 until the end of October that is one month before the eruption, a progressive eastward motion was observed. Interestingly, the eruption was preceded by 11 months of an increasing number of long period earthquakes (Mavonga *et al.* 2010). Between 2006 October 31 and December 5 (dates of last acquisition before and first acquisition after the onset of the eruption), the point P6 experienced an additional eastward displacements of 5–6 cm (Fig. 9b). Note that the Goma Volcano Observatory reported a significant seismic swarm that started one month before the eruption (BGVN 2007). This co-eruptive eastward step displacement also

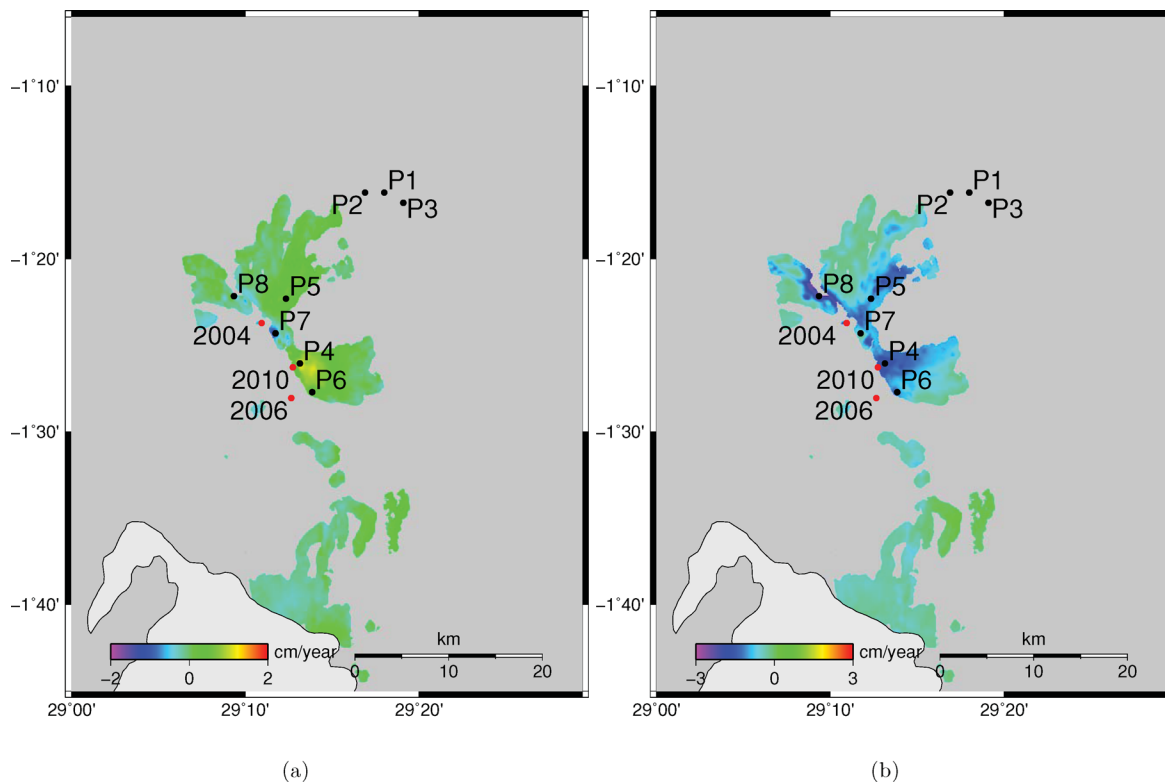


Figure 7. 2003–2010 east–west (left-hand panel) and vertical (right-hand panel) linear deformation rates calculated from all eight data sets. For points P4–P8 we present time-series below. Five small stable regions not shown here were used as reference during calculation of 2003–2010 linear deformation rates. 2004, 2006 and 2010 eruptive centres are shown as red points.

marks the beginning of a long-term westward drift at a rate of about -1 cm yr^{-1} that remains visible until the end of the observation period in 2010 September (Figs 8g–i and Fig. 9b). A co-eruptive uplift of about 1–2 cm is also detected (Figs 8g–i), but the most striking feature is the change of the long-term subsidence rate from less than a -1 cm yr^{-1} to about -2 cm yr^{-1} . Both, subsidence and horizontal westward movements triggered by the eruption are ongoing.

As in the case of the 2004 eruption, ground deformations are also observed within the main crater. From early May in 2006 (see star and arrow in Fig. 9e) point P7, located close to the NW caldera rim, experienced about -3 cm of subsidence that gradually reversed to the uplift two months later. Using conventional time-series methods based on a single acquisition geometry, Wauthier (2011) interpreted a 6-month-long signal (from 2006 May–November) as a possible pre-eruptive displacement. However, not all long-duration movements lead to an eruption and hence they cannot unambiguously be classified as precursors.

The 2010 January 2–27 eruption

Unlike the 2006 eruption, the 2010 eruption did not produce changes in the long-term displacement rate of points located to the NNE of the fracture (neither vertical nor horizontal).

Point P4, located close to the 2010 eruptive fracture along the S–SE flank, experienced 3 cm eastward displacement during the previous 2006 eruption and an additional 11 cm eastward co-eruptive displacement during the 2010 eruption (Figs 8a–c). Among these 11 cm, at least 3 cm started before the eruption, sometime between

2009 December 17 and 29 (see double stars and arrow in Fig. 9c). Deformation at other points located closer to the eruptive site (not shown here) suggest that the horizontal movement may have initiated earlier, between December 15 and 17. At the same time, an unambiguous pre-eruptive uplift of 3–4 cm started as early as between December 10 and 15 (see single star and arrow in Fig. 9c). For precise estimation of the pre-eruptive displacements we subtracted from time-series a linear trend computed for the 2007 June–2009 June time period. On these detrended time-series (not shown here) we observed that vertical uplift of P7 reached 2.9, 3.3 and 5.3 cm on 2009 December 15, 17 and 29, which is about 4 to 6 times larger than the rms calculated for this particular portion of time-series. Similarly, eastward displacement of P7 on the detrended time-series reached 4.7 cm on December 29, which is more than 4 times the rms. This unambiguous precursory signal, for the first time ever detected in VVP by three successive satellite images acquired by three different sensors in different geometries (ENVISAT, RS2 and ALOS) started more than two weeks before the onset of the eruption. This is particularly interesting as the seismicity only showed precursory activity less than 1 hr prior the lava outburst (BGVN 2010).

As for the two previous eruptions, deformation is also detected within the caldera. Note that the eruption started by a small fountaining activity within the crater. Point P7 moved -3 to -5 cm westward and subsided by about -3 cm . As along the eruptive fracture, these movements started clearly before the eruption, sometime between December 17 and 29 (see star and arrow in Fig. 9f). Another point located on the eastern side of the crater (not plotted here) moved in the opposite direction with a 2 cm eastward movement and 3–4 cm uplift.

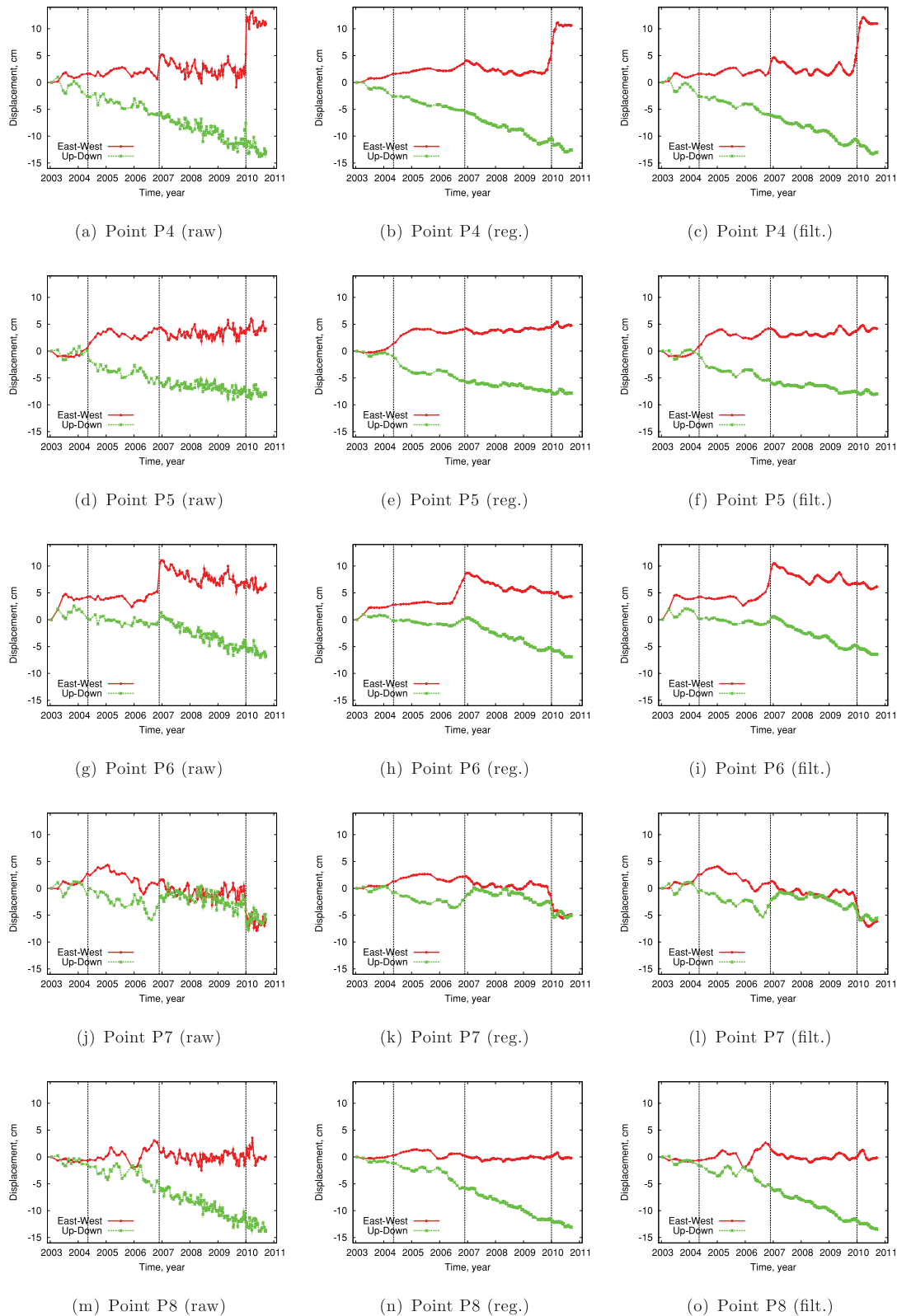


Figure 8. 2003–2010 east–west and vertical time-series of ground deformation at points P4–P8 (rows top to bottom). Raw (first column), regularized (second column) and filtered (third column) time-series are shown.

Table 3. Root mean square error and correlation coefficient comparing time-series from first (Fig. 6) and second (Fig. 8) runs for points P4, P6, P7 and P8 during 2006–2010. *—low correlation is observed due to absence of ground motion in east–west direction (see for example Fig. 8n).

Point/Processing	rms EW (cm)	Cor. EW	rms UP (cm)	Cor. UP
P4/Raw	1.1	0.95	0.9	0.92
P6/Raw	0.9	0.87	0.8	0.94
P7/Raw	0.9	0.94	0.8	0.91
P8/Raw	1.1	0.35*	0.9	0.94
P4/Regularized	0.5	0.98	0.5	0.97
P6/Regularized	0.5	0.97	0.5	0.97
P7/Regularized	0.6	0.98	0.5	0.96
P8/Regularized	0.7	0*	0.5	0.98
P4/Filtered	0.8	0.98	0.6	0.96
P6/Filtered	0.7	0.93	0.5	0.97
P7/Filtered	0.6	0.98	0.5	0.96
P8/Filtered	0.7	0.33*	0.5	0.97

5 CONCLUSIONS

In this paper we present a novel methodology for computing multidimensional time-series of ground deformation from SAR data acquired by multiple sensors with different parameters, such as azimuth and incidence angles, temporal and spatial sampling and resolution, wave-band, and polarization. Air-borne and space-borne data can be integrated, as well as ENVISAT data in Phase 3 (i.e. with varying azimuth). Raw, regularized and filtered time-series are presented and explained. Results of numerical simulation are shown in support of the validity of this approach.

The proposed method has the following advantages: (i) it achieves combined temporal coverage over an extended period of time when data from many different sensors with different temporal coverage are available; (ii) temporal resolution of produced time-series increases since it includes the combined sampling from all data sets, which helps to observe signal in more detail and also to improve the quality of post-processing (i.e. filtering); (iii) two or three components of ground deformation vector are computed, which helps in interpretation of observed ground deformation and further modelling and inversion; (iv) various sources of noise (i.e. tropospheric, ionospheric, topographic, orbital, thermal, etc.) are averaged out during the processing improving the signal-to-noise ratio. In addition our technique does not introduce any interpolation errors in comparison to methodology that separately solves ascending and descending data from different satellites using standard SBAS followed by interpolation of the velocity at uncommon data points. Interpolation error can become significant especially in case of temporarily sparse sets.

We apply presented methodology to studying ground deformation of the VVP, North Kivu, Democratic Republic of Congo during 2003–2010 and observed ground deformation with remarkable resolution and precision. We produce 2-D high-resolution time-series of lava compaction regions that show steady subsidence over the time of observation. We also observe long-term subsidence of Nyamuragira and 2004, 2006 and 2010 co-eruptive deformation. This signal is observed on both horizontal and vertical time-series. Finally we observe unambiguous pre-eruptive signals associated with the 2010 eruption of Mt Nyamuragira. Sharp signals of the amplitude up to 4–6 times the rms were observed approximate two

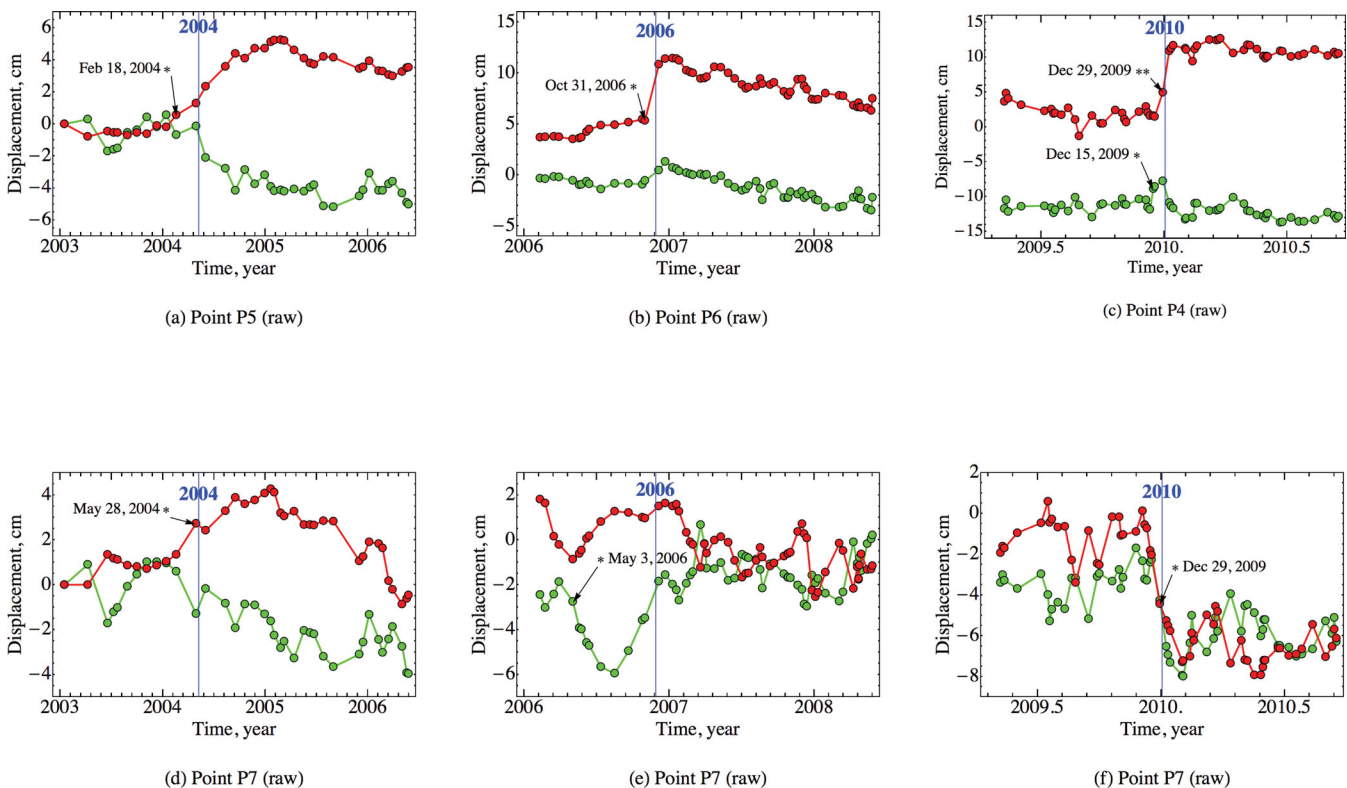


Figure 9. Close up to co eruptive deformations associated to 2004 (first column), 2006 (second column) and 2010 (third column) eruptions. Horizontal (red) and vertical (green) displacements observed close to corresponding eruptive fractures (top row) and within crater (bottom row). Vertical blue lines are onset of eruptions. See text for explanation of dates marked by stars and arrows.

weeks prior the 2010 eruption. They contrast with longer period (a few months long) signals that not necessarily lead to an eruption. Modelling of observed deformation is beyond the scope of this paper and will be presented in more detail in follow up studies.

Further studies will be performed for the region with dense GPS that can be integrated in the processing and used for cross validation and estimation of accuracy. We also intend to introduce a weighting factor to the eqs (3) and (4) based on sensor wave-band.

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REFERENCES

- Amelung, F., Jonsson, S., Zebker, H. & Segall, P., 2000. Widespread uplift and “trapdoor” faulting on Galapagos volcanoes observed with radar interferometry, *Nature*, **407**, 993–996, doi:10.1038/35039604.
- Beavan, R.J., Samsonov, S., Denys, P., Sutherland, R., Palmer, N.G. & Denham, M., 2010. Oblique slip on the Puysegur subduction interface in the 2009 July Mw 7.8 Dusky Sound earthquake from GPS and InSAR observations: implications for the tectonics of southwestern New Zealand, *Geophys. J. Int.*, **183**(3), 1265–1286, doi:10.1111/j.1365-246X.2010.04798.x.
- Berardino, P., Fornaro, G. & Lanari, R., 2002. A new algorithm for surface deformation monitoring based on small baseline differential SAR interferograms, *IEEE Trans. Geosci. Remote Sens.*, **40**(11), 2375–2383.
- BGVN. 2004. Bulletin Global Volcanism Network, Smithsonian Institution, Nyamulagira BGVN 04/2004 (29:04).
- BGVN. 2007. Bulletin Global Volcanism Network, Smithsonian Institution, Nyamulagira BGVN 01/2007 (32:01).
- BGVN. 2010. Bulletin Global Volcanism Network, Smithsonian Institution, Nyamulagira BGVN 08/2010 (35:08).
- Biggs, J., Lu, Z., Fournier, T. & Freymueller, J., 2010. Magma flux at Okmok Volcano, Alaska from a joint inversion of continuous GPS, campaign GPS and InSAR, *J. geophys. Res.*, **115**, B12 401, doi:10.1029/2010JB007577.
- Chorowicz, J., 2005. The East African rift system, *J. Afr. Earth Sci.*, **43**(1–3), 379–410.
- Colclough, S., 2005. Investigations of Nyamuragira and Nyiragongo volcanoes (DRC), using interferometric synthetic aperture radar, *PhD thesis*, University of Cambridge, Cambridge.
- d’Oreye, N. & Celli, G., 2010. Automatic InSAR systematic processing and web based tool for efficient data mining: application to volcano monitoring in Africa, *Cahier du Centre Europ. de Geodyn. et de Seismologie*, **29**, 23–32.
- d’Oreye, N. *et al.*, 2008. Systematic InSAR monitoring of African Active Volcanic Zones: what we have learned in three years, or a harvest beyond our expectations, in *Proceedings of the Second workshop on USE of Remote Sensing Techniques for Monitoring Volcanoes and Seismogenic Areas (USEReST 2008)*, Naples, pp. 57–62.
- Durieux, J., 2004. Nyiragongo: the January 10th 1977 eruption, *Acta Vulcanologica*, **14**(15(1–2)), 145–148.
- Ebinger, C., 1989. Geometric and kinematic development of border faults and accommodation zones, Kivu-Rusizi rift, Africa, *Tectonics*, **8**(1), 117–133.
- Farr, T. & Kobrick, M., 2000. Shuttle Radar Topography Mission produces a wealth of data, *EOS, Trans. Am. geophys. Un.*, **81**, 583–585.
- Fernandez, J. *et al.*, 2009. Gravity-driven deformation of Tenerife measured by InSAR time series analysis, *Geophys. Res. Lett.*, **36**, L04 306, doi:10.1029/2008GL036920.
- Gonzalez, P. & Fernandez, J., 2011. Drought-driven transient aquifer compaction imaged using multitemporal satellite radar interferometry, *Geology*, **39**(6), 551–554.
- Gonzalez, R.C. & Woods, R.E., 2001. *Digital Image Processing*. Prentice Hall, Upper Saddle River, NJ.
- Hansen, P. & O’Leary, D., 1993. The use of the L-curve in the regularization of discrete ill-posed problems, *SIAM J. Sci. Comput.*, **14**(6), 1487–1503.
- Komorowski, J.-C. *et al.*, 2004. The January 2002 flank eruption of Nyiragongo volcano (Democratic Republic of Congo): chronology, evidence for a tectonic rift trigger, and impact of lava flows on the city of Goma, *Acta Vulcanologica*, **14**(15(1–2)), 127–161.
- Lanari, R., Mora, O., Manunta, M., Mallorqui, J.J., Berardino, P. & Sansosti, E., 2004. A small-baseline approach for investigating deformations on full-resolution differential SAR interferograms, *IEEE Trans. Geosci. Remote Sens.*, **42**(7), 1377–1386.
- Lu, Z., Masterlark, T. & Dzurisin, D., 2005. Interferometric synthetic aperture radar study of Okmok volcano, Alaska, 1992–2003: magma supply dynamics and postemplacement lava flow deformation, *J. geophys. Res.*, **110**, B02403, doi:10.1029/2004JB003148.
- Lu, Z., Dzurisin, D., Biggs, J., Wicks, C., Jr. & McNutt, S., 2010. Ground surface deformation patterns, magma supply, and magma storage at Okmok volcano, Alaska, from InSAR analysis: 1. Interruption deformation, 1997–2008, *J. geophys. Res.*, **115**, B00B02, doi:10.1029/2009JB006969.
- Manzo, M. *et al.*, 2006. Surface deformation analysis in the Ischia Island (Italy) based on spaceborne radar interferometry, *J. Volcanol. Geotherm. Res.*, **151**, 399–416.
- Mavonga, T., Kavotha, S.K., Lukaya, N., Etoy, O. & Durieux, J., 2006. Seismic activity prior to the May 8, 2004 eruption of volcano Nyamuragira, Western Rift Valley of Africa, *J. Volcanol. Geotherm. Res.*, **158**, 355–360, doi:10.1016/j.jvolgeores.2006.06.021.
- Mavonga, T., Kavotha, S., Lukaya, N. & Etoy, O., 2010. Some aspect of seismicity prior to the 27 November 2006 eruption of Nyamuragira volcano and its implication for volcano monitoring and risk mitigation in the Virunga Volcanic Province, *J. Afr. Earth Sci.*, **58**, 829–832, doi:10.1016/j.jafrearsci.2010.02.002.
- Ozawa, T. & Ueda, H., 2011. Advanced InSAR time series analysis using interferograms of multiple-orbit tracks: a case study on Miyake-jima, *J. geophys. Res.*, doi:10.1029/2011JB008489.
- Peltier, A., Bianchi, M., Kaminski, E., Komorowski, J.-C., Rucci, A. & Staudacher, T., 2010. PSInSAR as a new tool to monitor pre-eruptive volcano ground deformation: validation using GPS measurements on Piton de la Fournaise, *Geophys. Res. Lett.*, **37**(12), L12301, doi:10.1029/2010GL043846.
- Press, W.H., Teukolsky, S.A., Vetterling, W.T. & Flannery, B.P., 2007. *Numerical Recipes in C: The Art of Scientific Computing*, 3rd edn, Cambridge University Press, Cambridge, 994pp.
- Samieie-Esfahany, S., Hanssen, R., van Thienen-Visser, K. & Muntendam-Bos, A., 2009. On the effect of horizontal deformation on InSAR subsidence estimates, in *Proceedings of the Fringe 2009 Workshop, ESA SP-677, ESTEC, Noordwijk, The Netherlands*.
- Samsonov, S., Tiampo, K., Gonzalez, P., Manville, V. & Jolly, G., 2010. Modelling deformation occurring in the city of Auckland, New Zealand mapped by the Differential Synthetic Aperture Radar, *J. geophys. Res.*, **115**, B08410, doi:10.1029/2009JB006806.
- Samsonov, S., Beavan, J., Gonzalez, P., Tiampo, K. & Fernandez, J., 2011. Ground deformation in the Taupo Volcanic Zone, New Zealand observed by ALOS PALSAR interferometry, *Geophys. J. Int.*, **187**(1), 147–160.
- Smets, B., Wauthier, C. & d’Oreye, N., 2010. A new map of the lava flow field of Nyamulagira (D.R.Congo) from satellite imagery, *J. Afr. Earth Sci.*, **58**(5), 778–786.

- Stevens, N., Wadge, G., Williams, C., Morley, J., Muller, J.P., Murray, J. & Upton, M., 2001. Surface movements of emplaced lava flows measured by synthetic aperture radar interferometry, *J. geophys. Res.*, **106**(B6), 11293–11313.
- Stramondo, S., Chini, M., Bignami, C., Salvi, S. & Atzori, S., 2011. X-, C-, and L-band DInSAR investigation of the April 6, 2009, Abruzzi earthquake, *IEEE Geosci. Remote Sens. Lett.*, **8**(1), 49–53, doi:10.1109/LGRS.2010.2051015.
- Tazieff, H., 1977. An exceptional eruption: Mt. Nyiragongo, January 10th. 1977, *Bull. Volcanol.*, **40**(3), 1–12.
- Tedesco, D. *et al.*, 2007. January 2002 volcano-tectonic eruption of Nyiragongo volcano, Democratic Republic of Congo. *J. geophys. Res.*, **112**, B09202.
- Tikhonov, A.N. & Arsenin, V.Y., 1977. *Solution of Ill-Posed Problems*, V. H. Winston and Sons, Washington, DC.
- Usai, S., 2003. A least squares database approach for sar interferometric data, *IEEE Trans. Geosci. Remote Sens.*, **41**(4), 753–760.
- Wauthier, C., 2011. InSAR applied to the study of active volcanic and seismic areas in Africa, *PhD thesis*, Liege University.
- Wauthier, C., Cayol, V., Kervyn, F. & d'Oreye, N., 2012. Magma sources involved in the 2002 Nyiragongo eruption, as inferred from an InSAR analysis, *J. geophys. Res.*, **117**, B05411, doi:10.1029/2011JB008257.
- Wegmuller, U. & Werner, C., 1997. Gamma SAR processor and interferometry software, in *Proceedings of the Third ERS Symposium on Space at the service of our Environment*, Florence.
- Wei, M., Sandwell, D., Fialko, Y. & Bilham R., 2010. Slip on faults in the Imperial Valley triggered by the 4 April 2010 Mw 7.2 El Mayor-Cucapah earthquake revealed by InSAR, *Geophys. Res. Lett.*, **38**, L01308, doi:10.1029/2010GL045235.
- Wright, T., Fielding, E. & Parson, B., 2001. Triggered slip: observations of the 17 August 1999 Izmit (Turkey) earthquake using radar interferometry, *Geophys. Res. Lett.*, **28**(6), 1079–1082.
- Wright, T.J., Parsons, B.E. & Lu, Z., 2004. Toward mapping surface deformation in three dimensions using InSAR, *Geophys. Res. Lett.*, **31**, L01607, doi:10.1029/2003GL018827.